

NAME

Toronto, Ontario | email | number | [GitHub](#) | [LinkedIn](#)
AI/ML & Data Science | Open Source Contributions - Pandas

EDUCATION

BSc (Hons.) Computer Science Expected: May 2027
York University, Toronto, Ontario
Advanced Math(Calculus, 3D Geometry, Probability, Statistics), Programming(Python, SQL, Java, C), Data Structures and Algorithms, Artificial Intelligence, Relational databases, Risk Management

EXPERIENCE

Research — AI and Machine Learning, University Sep.'25-Dec.'25
➤ Designed a Reinforcement Learning system using PyTorch, Sentence Transformers, and FastAPI.
➤ Applied NLP and transformer based reasoning to guide RL rewards, improving code generation accuracy by 15% on 400+ competitive programming problems.
➤ Analyzed 300+ GitHub logs in Microsoft Excel for faculty research, identifying software bugs.
➤ Preparing academic submission at ACM International Conference 2026, Montreal.

PROJECTS

Open-Source Contributor — Pandas (10 Million+ global users)
➤ Bug fix and added test cases for *ExtensionArray* operations; 15+ commits with multiple code reviews. (PR [merged](#))
➤ Improved CSV parsing with *on_bad_lines*; contributed to iterative development, unit testing and debugging.

Demand Forecasting and Anomaly Detection Dashboard [GitHub](#) May 2025
➤ Processed 40M+ NYC taxi trips, cleaned 1.2M+ outliers, and added new features using PySpark, Dask, and Pandas.
➤ Trained predictive models - XGBoost (13% MAPE) for demand and LSTM (0.055 MSLE) for detecting anomalies.
➤ Built a Power BI dashboard from model outputs stored in MySQL for easy visualization and analysis.
Data Engineering, Power BI, SQL, Python, XGBoost, LSTM , PySpark, Dask, Pandas, Data Mining, Tensorflow

Quantitative Trading Algorithm with LSTM [GitHub](#) July 2025
➤ Built an ensemble trading model with LSTM, Bollinger Bands, and RSI, executing trades when 2+ indicators aligned.
➤ Achieved +54.6% cumulative return and 6.5% CAGR over 5 years across 50+ stocks.
➤ Outperformed the S&P 500 by 17%, achieved a 57% return in the best year, and preserved capital during COVID-19 in volatile market conditions.
Time Series Forecasting, Feature Engineering, Statistical Analysis, LSTM, Pandas, NumPy, scikit-learn, TensorFlow, Financial Analysis, Data Visualization (seaborn, quantstats), Data Modeling

Azure Deployed Emotion Detection with Docker [GitHub](#) March 2025
➤ Trained a ResNet-based CNN with L2 regularization and batch norm, achieving 75% accuracy on FER2013 (Kaggle).
➤ Integrated MLflow to monitor model performance and data quality.
➤ Containerized with Docker and deployed on Azure, featuring a Streamlit + WebRTC app.
Computer Vision, MLflow, Docker, Azure, Python, TensorFlow (Keras API), OpenCV, CNNs, Data Collection

AI RAG Assistant Using LangChain + LLMs [GitHub](#) September 2025
➤ Built a context-aware chatbot using LangChain and Watsonx LLM to provide answers from documents.
➤ Created a document data pipeline with RAG that splits PDFs into 100-token chunks.
IBM Watsonx LLM, LangChain, Retrieval-Augmented Generation (RAG), Semantic Search, Prompt Engineering

SKILLS

Machine Learning Algorithms: Regression, Classification, Clustering, CNNs, LSTM, Reinforcement Learning
LLM & NLP Frameworks: IBM Watsonx, LangChain, LangGraph, Hugging Face, RAG, Open AI API
Libraries: scikit-learn, TensorFlow, Keras, PyTorch, NumPy, Pandas, PySpark, XGBoost, JAX, Matplotlib, Plotly
Languages: Python, SQL, R, Scala, Java
Cloud & DevOps: Docker, Kubernetes, Azure, GCP, CI/CD, FastAPI, Linux, Unix, MLflow, Git, Github
Other tools: Databricks, Datameer, Teradata, Tableau, Excel, PowerBI

CERTIFICATIONS

IBM AI Engineering Professional Certificate
Built and deployed Generative AI and Deep Learning models with end-to-end ML workflows, including data analytics, preprocessing, feature engineering, model training, evaluation and cloud deployment workflows.